

Integration of Monte Carlo Transport Simulations into a TCAD Workflow for Electron Detector Developments

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Abstract—A new methodology is presented and allows for the integration of Monte Carlo electron distributions into the Sentaurus workflow for the development of electron detector applied to transmission microscopes. The proposed method consists in converting electron positions generated by the Monte Carlo software in an optical generation file, which can be used in Sentaurus device simulations. Results are validated by means of comparisons between electron distribution provided by the two simulators, and by a comparison with an experiment. Limitations are also discussed.

Index Terms—Image sensors, CMOS image sensors, Electron beams, CMOS, Simulation, Monte Carlo methods.

I. INTRODUCTION

TRANSMISSION electron microscopes (TEM) use more and more direct electron detectors in order to record pictures [1], [2]. These detectors are directly exposed to the electron beam and does not require a scintillator previously employed for doing the conversion electron-photon. Developments of such detectors can be quite tricky, as they have to face a harsh environment composed of radiations inducing strong ionization doses [3], and because the large spread of the electron distribution in the detector leads to compromises between the sensitivity and the spatial resolution [4], [5]. Over the last ten years, the development of direct electron detectors [6], [7] and image processing software has enabled spectacular advances in techniques such as the cryo-microscopy [8]. In order to protect the sample from damage caused by electron radiation, it is imperative to work with a low-dose beam and therefore to have excellent detection quality from the detectors used. Thus, detectors have been developed with a large pixel pitch for an improve sensitivity but only compatible with a low energy beam (100 keV) [9]. However, conventional beam energies such as at 200 keV are necessary to maintain good spatial resolution performances. Therefore, electron detectors need to be improved and especially their sensitivity which has to be much higher compared to current developments, while maintaining high spatial resolution [10].

TCAD simulations are widely used for semiconductor device developments and in particular for complementary metal oxyde semiconductor (CMOS) image sensors. Thus, main

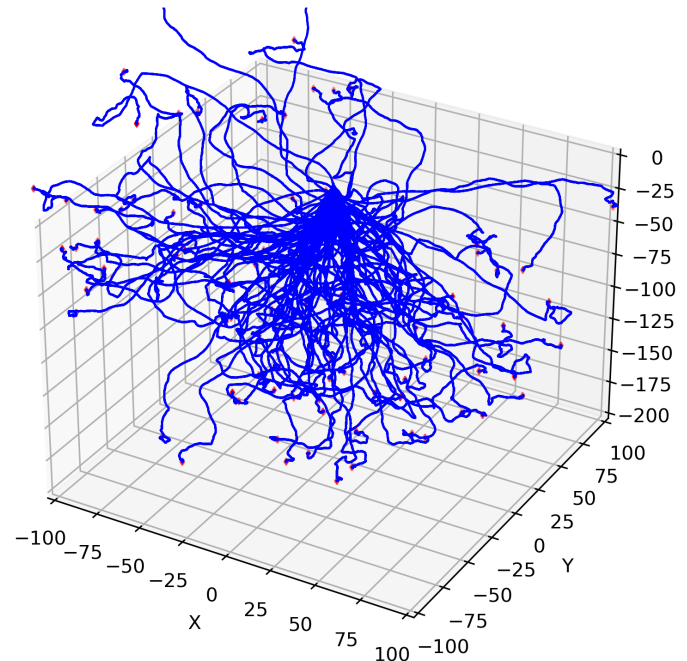


Fig. 1. 3D distribution of primary (red points) and secondary (blue points) electrons generated by a 200 keV electron beam in a silicon substrate, performed by means of the Casino Monte Carlo software with 100 incoming electrons. Dimensions are in μm .

TCAD softwares generally include options for light illuminations of the simulated device with the aim of modeling the electron hole pairs distribution created in the photo-detector by the light source [11], [12]. However, no native solution is proposed in TCAD software to simulate electrons and more generally particles distributions created by a beam or an accelerator into an image sensor. Actually, results of Geant4 simulations can be adapted to calibrate heavy ion or alpha particles TCAD parameters [13]–[15], or TCAD simulations can be combined with a Monte-Carlo software output to a third party simulator [16]–[18]. In general, these simulation frameworks are used to study ionization tracks of ions, the model alone proposed in TCAD tools showing practical limits [15]. Consequently, there is a need for proposing a method directly including the simulated electron-hole distribution created by particles with a Monte-Carlo software into the TCAD tool, in order to fully benefit from the TCAD ability to simulate complex carrier diffusions in biased devices. Indeed, the collecting node or the photodiode potential is floating during the

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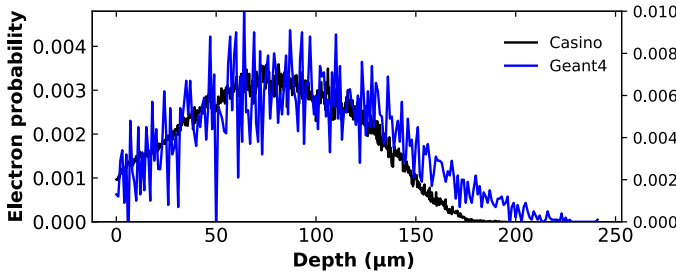


Fig. 2. Comparison of electron distribution generated at 200 keV for 10000 incoming electrons with Casino and Geant4. The substrate is a bulk silicon. The Geant4 *em standard opt4* is used for this simulation.

integration and therefore varies with the amount of integrated carriers. In addition, gates might be turned on or off during the pixel operation, and all these electrical variations must be simulated in order to precisely reproduce the detector performances. This feature is not possible with the use of a third party simulator [18] and only available in the TCAD software. In addition, particle Monte Carlo modeling can give an accurate volume deposition for each electron tracks, where as analytical formula can give an average value of the deposition. As shown in Fig. 1, some electrons can reach high distance from the main track of the beam and this should be taken into account for detector developments.

The aim of this work is therefore to propose a new method in order to integrate the electron hole pair distribution simulated by the Monte Carlo Casino software [19] into the TCAD workflow, thanks to a script executed in the TCAD environment. The final goal being to simulate direct electron detectors for TEM, the Monte-Carlo Casino software is chosen because it is especially developed for electron beam simulations [19], and widely used for simulating scanning electron microscope [20], [21], TEM [22], [23] experiments or equipment with beam energies up to 300 kV [24]. Fig. 1 is showing an example of Casino simulation for a 200 kV electron beam, based on 100 incoming electrons into a silicon substrate. Casino gives all primary and secondary electrons positions in the material. The goal is to convert this information into optical generated carriers readable by the TCAD software. The Geant4 platform [25]–[27] might be an alternative to Casino, and the proposed method developed in this article should be usable with Geant4 as well. The latter one is also used for TEM detector simulations [28] and Fig. 2 is showing that both softwares lead to comparable results in the first 150 μm of interaction in silicon.

After a presentation and the description of the method, comparisons with Monte Carlo distributions and experimental results are performed in order to validate the proposal.

II. DESCRIPTION OF THE METHODOLOGY

In this work, TCAD simulations are performed with the Synopsys Sentaurus 2020-09 software, in two dimensions. The device is reproduced with the Sentaurus device editor (SDE) tool, by means of secondary ion mass spectroscopy (SIMS) profiles and information about the pixel design which has a 8.5 μm pitch [5] and is based on three transistors. The detector

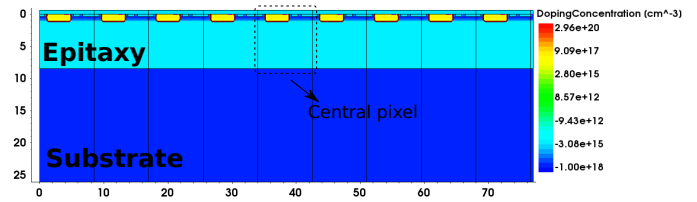


Fig. 3. Two dimensional (2D) TCAD simulation of the doping profile of the structure needed for the MTF extraction. 9 pixels are simulated, and one ground contact is put at the right extremity of the structure.

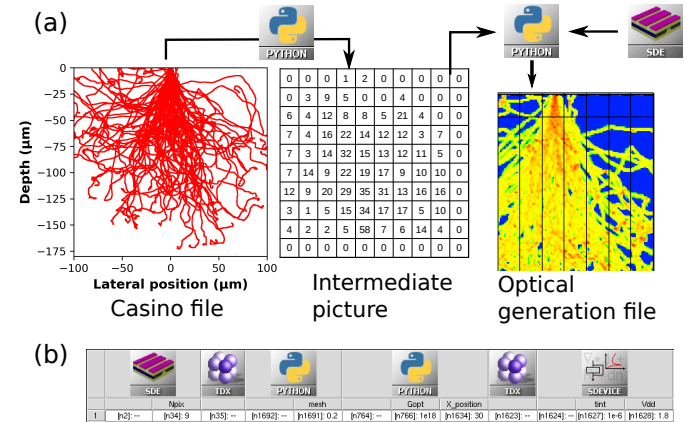


Fig. 4. (a) Schematic overview of the conversion flow from Casino to Sentaurus. (b) integration in the Sentaurus work-bench of the conversion steps.

including 1280×780 of a such pixel has been characterized in a TEM and the modulation transfer function (MTF) of the detector is measured and compared with simulations later in this paper. For the need of the MTF extraction, a point spread function (PSF) is simulated in TCAD thanks to a structure based on 9 pixels where the electron beam is moved from one extremity to the other one, the integrated signal being monitored in the central pixel (see Fig. 3). The Fourier transform of the resulting curve gives the MTF. It is therefore mandatory to perfectly emulate the shape of electrons generated in the substrate by the microscope beam. Each pixel has a floating node initialized by a reset transistor simulated in mixed mode.

A. Overview of the proposed method

The Casino output is a data file including all the positions and various cascades of electrons generated by primary electrons. A script can then be written with the goal of extracting all the primary and secondary electrons coordinates and the particle positions can be plotted for instance as in Fig. 1. However, these data are not compatible with Sentaurus inputs.

The idea is to convert the various electrons positions into an optical generation data. For this purpose, the conversion is performed in two steps:

- creation of an intermediate picture with a uniform mesh, where each point contains a number of generated electrons at the grid point
- read the Sentaurus grid file, and for each grid point interpolate the expected number of generated electrons

from the intermediate picture. A new *dat* file is written containing the optical generation and is converted to the *tdr* format. The script is provided in this paper.

Before executing this script, the Sentaurus grid file in binary format needs to be converted in DF-ISE files in order to be readable by an external editor, which is performed by the TDX tool available in the Sentaurus workflow. Then, a script written in Python and based on the Kelzenberg work initially done for a Lumerical to Sentaurus conversion [29] allows for the grid reading and an extraction of all regions names where the optical generation is performed. For the purpose of facilitating this task, a special user field is introduced during the structure generation by SDE, under the name *PMIUserField0* and *PMIUserField1*, as explained by Kelzenberg in his work [29]. By doing so, the region names, the axis numbers and the data point numbers where the optical generation is introduced are identified and the user field is replaced by the optical generation field during the data file writing operation.

The Casino to Sentaurus conversion is performed in two steps because scanning all particle coordinates for each Sentaurus grid point would require a much longer process time. Indeed, with the proposed method, the particle coordinates are scanned only once, then the script directly reads the interpolated value in the intermediate picture at the requested coordinate. All these steps can be executed in the Sentaurus work-bench and an overview of the conversion flow is shown by Fig. 4.

B. Construction of the intermediate picture

The intermediate picture has a fixed mesh in lateral and in vertical dimension, and the resolution grid should be in the range of the Sentaurus epitaxy largest resolution in lateral and depth directions. Making a uniform mesh simplifies the process but it is supposed that it does not impact the final electron distribution seen in Sentaurus. Indeed, in the substrate region where the Sentaurus mesh is larger, the final optical generation might miss some particles as not all intermediate picture points are read. On the other hand, near the surface where the Sentaurus mesh is much smaller, the final particle distribution will lose some precisions on individual particle position and on the main electron distribution borderline. As the error is defined by the intermediate picture mesh, its impact should be limited with a resolution in the order of several 100 nm. The lateral dimension of the intermediate picture has to be larger than the device one, and the depth is chosen to be larger than the deepest position of the generated electrons. An array is then created and the script calculates for each particle the picture coordinates corresponding to the particle position, and the picture element is incremented accordingly. At the end, the picture array is multiplied by an optical generation factor with the aim of approaching the requested carrier generation number in the Sentaurus device simulation (see the Fig. 4). Indeed, the optical generation in Sentaurus is given in $\text{cm}^{-3}\text{s}^{-1}$ so the numbers in the intermediate picture in electrons need to be converted and also adapted with the expected number of incoming electrons during the integration time or the expected integrated electrons in the pixel.

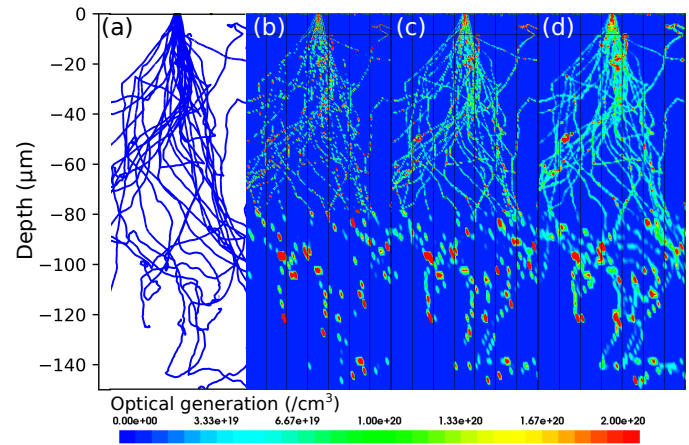


Fig. 5. Comparison of electron distribution generated by 30 electrons accelerated at 200 kV, from Casino (a), in Sentaurus with the following resolution in the intermediate picture: 100 nm (b), 200 nm (c) and 500 nm (d). The color scale refers to the optical generation in Sentaurus, with an optical generation factor equal to 2.5×10^{19} .

C. Creation of the optical generation file

Once all region names of the Sentaurus grid file have been read, a script takes all grid elements of each region name and calculates the corresponding optical generation. The first step is to create an array with identical dimensions compared to the intermediate picture which contains the coordinates of all picture elements given in μm . The lateral coordinates are eventually shifted in order to take into account the incoming beam position on the substrate. Then, knowing the Sentaurus grid element coordinates, the optical generation is interpolated from the values of the intermediate picture at the surrounded coordinates. This operation is performed by the Python *interp* function from the *scipy* library, giving the multidimensional interpolation on rectilinear grids. The *dat* file is then re-written with the optical generation data, and the file is converted into the *tdr* format.

III. VALIDATION OF THE METHODOLOGY

In this section the output of one Casino simulation is compared with the Sentaurus optical generation file, and the distribution of generated electrons at various substrate depths are analysed with the aim to check if the simplification consisting in using a uniform mesh on the intermediate picture does not lead to improper charge numbers or distribution.

30 electrons accelerated at 200 kV are simulated with Casino, by using a silicon substrate of $60 \times 200 \mu\text{m}^2$ with a top silicon oxide of $5 \mu\text{m}$ similarly to the simulated TCAD device. Fig. 5 is showing the resulting electron distribution in the silicon substrate together with Sentaurus optical generation data obtained for the following resolutions in the intermediate picture: 100 nm, 200 nm and 500 nm. The Sentaurus device simulation uses the device shown in Fig. 3 with a beam position set at $30 \mu\text{m}$ which is approximately the middle of the device. In the first $10 \mu\text{m}$ depth where the mesh resolution is maximum 100 nm, the electron distribution is more accurate with the smallest intermediate picture resolution. This observation is especially visible on the magnification provided by

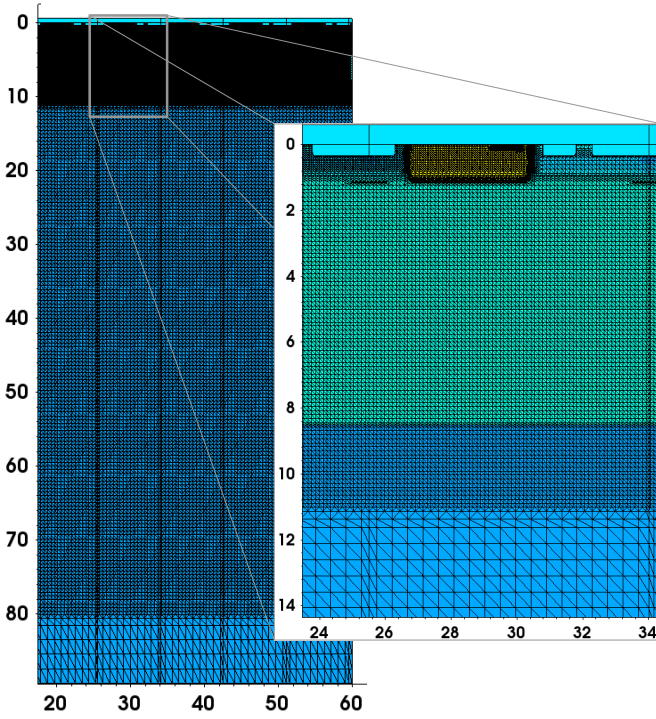


Fig. 6. doping distribution of the simulated device with the mesh in dark lines.

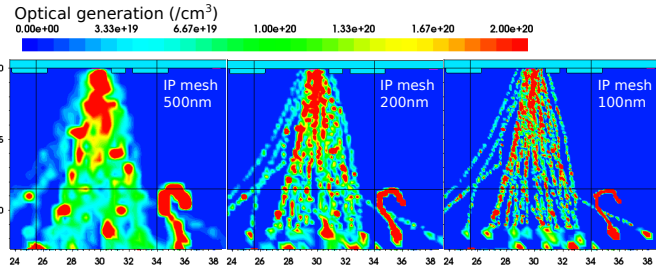


Fig. 7. Comparison of electron distribution generated in Sentaurus of the three intermediate picture (IP) meshes. The unit is interpreted as optical generation in Sentaurus.

Fig. 7. However, the Sentaurus mesh resolution is increased to 500 nm from 10 μm to 80 μm , and to 2 μm from 80 μm (Fig. 6), and the case where the intermediate picture mesh is small results in missing particles (see the case (b) in Fig. 5). This issue is due to the fact that Sentaurus grid points are more spaced than the intermediate picture ones, and therefore some data of the intermediate picture are not read. Therefore, if a depth larger than 10 μm is considered, an intermediate picture resolution larger than 200 nm is more favourable. The choice for the intermediate picture resolution is therefore a compromise and in our case it is found that 200 nm is a good candidate, favouring an acceptable accuracy on the first 80 μm , where electrons are supposed to be attracted and captured by the photodiode.

It would have been possible to modify the script in order to manage two or more different mesh pitches for the intermediate picture, but it would require an adaptation of the optical generation factor for each mesh. Using only one mesh allows

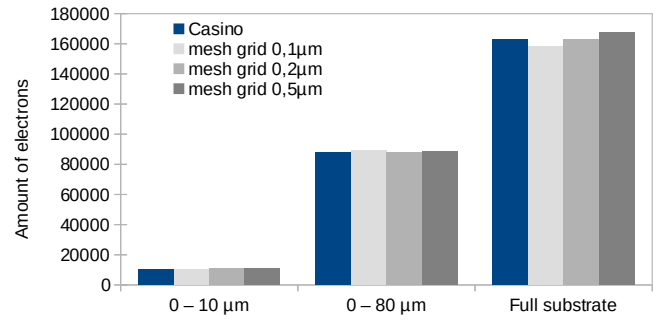


Fig. 8. Comparison of generated electrons in the substrate given by the Casino simulation for 30 incoming electrons at 200 kV, and by Sentaurus for three mesh grids in the intermediate picture: 100 nm, 200 nm and 500 nm. The integration time is fixed at 1 μs , and the optical generation factor has been adapted in Sentaurus for the three variations in order to approximately match the Casino result. The full substrate is 500 μm thick.

for a simple process, for which the conversion of the particle distribution could miss particles deeply in the substrate, in a region where electrons are recombined and do not influence the detector operation.

Last, the amount of electrons generated in the substrate for the four conditions of Fig. 5 and in 3 regions of the substrate is plotted in the Fig. 8:

- from the surface to 10 μm depth: region with the smallest mesh in Sentaurus
- from the surface to 10 μm depth: region including the area with a mesh pitch at 0.5 μm
- all the substrate: includes in addition to the previous ones the area with a mesh pitch at 2 μm

This simulation is performed with Sdevice, taking into account the optical generation of Fig. 5, and an illumination time of 1 μs . All contacts are put at 0 V as only the optical generation in the substrate is integrated. The amount of electrons generated in the full substrate might look small compared to the expected ionized electron number, calculated by means of the following expression:

$$MIP \times e_{in} \times z_{range} \quad (1)$$

where MIP is the minimum ionizing number (80 $\text{e}^-/\mu\text{m}$ [30]), e_{in} is the number of incoming electrons (30) and z_{range} is the range of the final electron distribution in depth (150 μm). Thus about 360000 electrons should be expected. Actually the amount of electrons shown in the full substrate of Fig. 8 is smaller due to the reduced dimension of the device inducing missing electrons in the final distribution. As can be seen, the amount of generated electrons remains similar, in all regions and for the three meshes of the intermediate picture. Thus, it demonstrates that the intermediate picture constant mesh does not lead to an over or under conversion of electrons in optical generation in the various regions. The only limitation of the constant mesh remains the eventual drop in particle position accuracy in regions where the Sentaurus mesh is much smaller.

IV. COMPARISON WITH MEASUREMENTS

In this part the proposed simulation method is used to extract the modulation transfer function (MTF) of the detector,

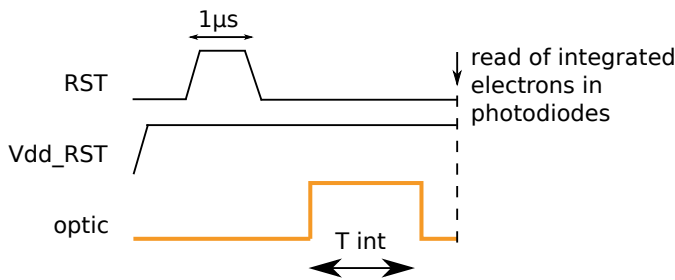


Fig. 9. Timing diagram used for the TCAD simulation of the point spread function. One simulation is needed for each beam position.

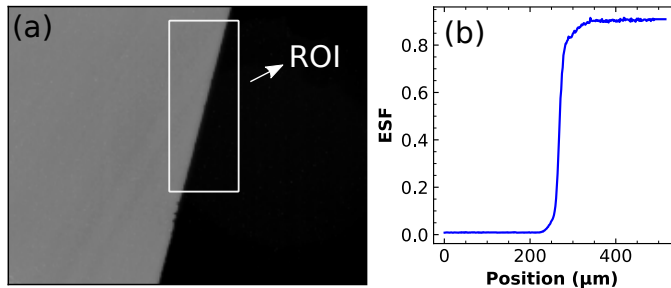


Fig. 10. (a) TEM picture of the slanted edge observed in the TEM with a 200 kV electron beam. A white rectangle shows the region of interest use for the ESF extraction. (b) Extraction of the edge spread function.

and it is compared with an experimental result.

As written before, the TCAD simulation of the MTF is obtained by means of the Fourier transform of the point spread function simulated by the movement of the electron beam from one side to the other side of the 9 pixel structure shown in Fig. 3. 9 pixels is enough in order to correctly simulate the MTF, as the simulated point spread function reaches the asymptotic value of zero at the extremity of the structure. 10 beam positions per pixel are simulated and the integration time is fixed at $0.1 \mu\text{s}$. The optical generation factor is chosen in order to fill half of the pixel full well capacity, and up to 1.3×10^6 electrons are generated in the structure. The floating diffusion of each pixel is initialized with a reset transistor biased at 1.8 V , simulated in mixed mod [31], according to the timing diagram given in Fig. 9. The Casino simulation provides the electron distribution generated by a 200 kV beam which has a width of 10 nm , and 100 incoming electrons. The dielectric layers on top of the silicon substrate has been entered in the Casino accordingly to the design rule manual of the technology.

The experiment is conducted with a Hitachi HF2000 TEM at the CEMES-CNRS, the beam energy being fixed at 200 kV. The MTF of the detector is extracted by means of the slanted edge method [32], thanks to a tilted and opaque edge placed above the detector. A picture including the tilted screen together with a dark image and a flat field image (without any object) are recorded and a script extracts the edge spread function, as illustrated by Fig. 10. Then, a derivative of the edge spread function gives the line spread function, and the MTF is obtained by applying the Fourier transform of the latter one.

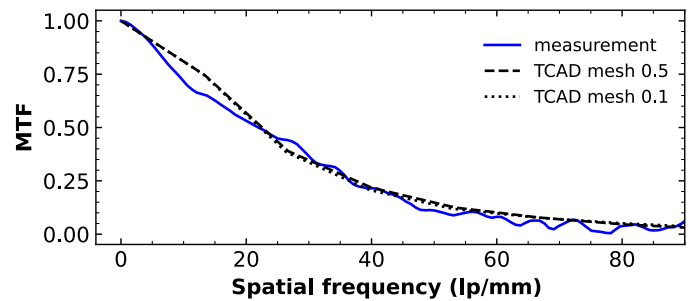


Fig. 11. Experimental measurement of the MTF with a 200 kV electron beam, and comparison with the TCAD simulations performed on the same detector with the inclusion of the Casino simulation of electron distribution. 'mesh' indicates the intermediate picture grid resolution.

The Fig. 11 shows the comparison between the experimental and the TCAD simulations performed with the proposed method for the MTF extraction. The Nyquist frequency is at 58.8 lp/mm and therefore the measured MTF is not very high. This result is expected as the detector is not especially designed for electron microscopy applications. It can be seen that the simulation curves follow well the experimental one, with some deviations potentially caused by experimental artefacts such as the edge quality of the slanted edge, and by the small number of pixels used in the TCAD simulation leading to less than 10 points in the MTF curve. Nevertheless, this good result validates the proposed method for the integration of the Monte Carlo Casino electron distribution in a Sentaurus workflow. The two different mesh grids of the intermediate picture do not influence the final result, and the MTF curve are almost superimposed.

V. EXTENSION TO 3D SIMULATIONS

The presented simulation flow is extended to a three dimension (3D) configuration, compatible with 3D TCAD Sentaurus simulations, in order to address conformation studies or any other kind of work requiring the three dimensions. For this purpose, the script generating the intermediate picture is modified, and now generates a 3D picture from the Casino output file. It is now needed to specify the three dimensions of the picture, and to choose the three mesh grids. The output file is then reshaped in 2 dimensions for saving in a text file format. Then, the script creating the optical generation file is also modified in order to manage the third dimension. After shaping back the third dimension of the intermediate picture file, the same interpolation function is used in order to estimate the optical generation on each point of the sentaurus mesh.

Fig. 12 is showing the result of a such a simulation. First, 100 electrons are implanted in a silicon bulk with Casino at 200 kV. Then, an intermediate picture is created with the dimensions $10 \times 10 \times 50 \mu\text{m}^3$, and a silicon substrate having a $8.5 \mu\text{m}$ pitch pixel is generated in Sentaurus Structure Editor. Finally, a 3D optical generation file is created from the intermediate picture and the 3D doping distribution, with a beam position at $x = 4 \mu\text{m}$ and $y = 4 \mu\text{m}$, and an optical generation factor of $10^{18} \text{ cm}^{-3} \cdot \text{s}^{-1}$. The cross-section extracted from the 3D TCAD simulation at $y = 4 \mu\text{m}$ is compared with a 400 nm thick cross-section extracted from the Casino

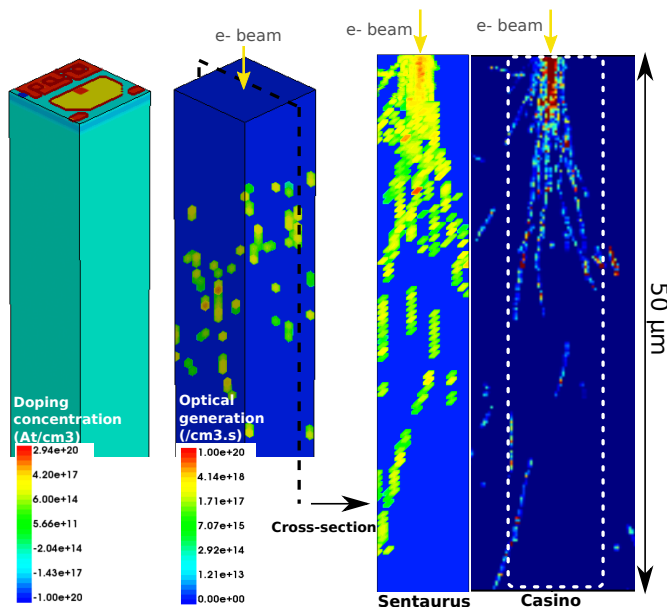


Fig. 12. On the left, 3D TCAD simulation of the doping concentration for a 8.5 μm pixel and 3D optical generation generated by a 200 kV electron beam. On the right, 2D TCAD cross section showing the optical generation distribution, and comparison with the Casino simulation output. The white dashed area represents the dimension of the TCAD simulation.

output at $y = 4 \mu\text{m}$; the optical generation on both cross-section looks comparable, validating the script. Inaccuracies or missing particles in the TCAD cross-section are due to the fact that the mesh in the substrate (every $1 \mu\text{m}$) is larger than the one of the intermediate picture ($0.2 \mu\text{m}$).

VI. DISCUSSION

The proposed methodology for integrating the Casino Monte Carlo particle distribution into the Sentaurus workflow is now validated and is discussed in this section. This flow may present some drawbacks and the first one is probably the choice of using only one grid for the intermediate picture which may lead to an inaccuracy or an underestimation of electron distributions in the final optical generation file. This is especially true for 3D simulations where the TCAD substrate mesh is much larger compared to the region near the photodiode and lead to missing charge in the substrate. However, it has been demonstrated that this issue does not have any visible effect on 2D simulations as the grid remains small enough. For 3D simulations, one possible improvement could be to use two different meshes for the intermediate picture. Another drawback is the fact that this protocol cannot be used to simulate a temporal particle propagation in the detector with the aim to follow its temporal track [33]. Only 2D fixed distributions can be imported in Sentaurus.

On the other hand, the execution of the Casino conversion flow is quite simple as it requires few parameters which are the intermediate picture dimensions and its mesh resolution. Moreover, the conversion process is fast. Indeed, on a computer equipped with a Xeon 2.1 GHz processor, the creation of the 2D intermediate picture used in part IV from 10^6 electron positions needs 1 mn and the conversion to the optical

generation requires 40 s, while the device structure has about 97000 points.

As a perspective, this methodology could be applied to Geant4 simulations, which could provide a wider usage of this work. Indeed, Geant4 is probably a better choice for particle distribution simulations as this software is largely used in the particle detector community.

VII. CONCLUSIONS

In this work a new methodology is proposed in order to integrate the Monte Carlo simulation output of electron distributions in the Sentaurus workflow. For this purpose, the conversion is performed in two steps by means of a python script fully integrated in Sentaurus. During the first step all particle positions are added in an intermediate picture defined by two dimensions and one mesh grid. Then, this picture is used to interpolate the Sentaurus optical generation at each mesh points. This process is validated with a comparison between the Casino electron distribution and the Sentaurus optical generation file, and also by means of a comparison between simulations and a measurement of a MTF. While this method is fast and simple, it has been extended to 3D simulations with some possible limitations related to the mesh difference between the pixel region and the substrate; however 2D TCAD device simulations usually provide enough information and are preferred for device investigations. The proposed method could be very useful for the development of future direct electron detectors.

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